FloodLens Pakistan

District heavy-rain warnings 1–3 days ahead and a weekly food-price watch · HackInsight 2026 · team Spark H

Problem and user

The 2025 monsoon killed more than 1,000 people in Pakistan (NDMA). District and PDMA planners need 24–72 hours of notice, per district, to decide where to pre-position teams and supplies and where to issue local warnings.

Approach

- **Model A (primary):** probability that a district's wettest-pixel rain exceeds its own 95th percentile of wet-day rain (NASA IMERG Late, 2015–2023), 1–3 days ahead, from archived weather-model forecasts (Open-Meteo Previous Runs) and observed rain lagged ≥14 h.
- **Model B (secondary, data ready):** weekly price change for 51 items in 17 cities (PBS SPI), 169 weeks collected.
- **Insight Agent:** answers from NDMA sitreps and advisories, quoting the exact section, with numbers inserted from tables with row IDs. It says "I don't know" when nothing matches, points emergencies to Rescue 1122 and NDMA/PDMA, and has Urdu output.

Pipeline

- Hourly Prefect flow: fetch new data → raw files + manifest (source, query, UTC time, SHA-256, licence) → rebuild all tables in a scratch DB → quality agent (fix/fill/flag, logged and replayed) → score → publish `status.json` for the dashboard.
- `make data` rebuilds everything from the frozen raw snapshot and refuses any file whose hash changed.
- Leakage tests: forecast inputs only from forecasts issued 1–3 days ahead; observed rain only from days that ended ≥14 h before issue. Fixed splits; the test set is locked in code.

Results: test set (Jan–Sep 2026, run once, frozen model) and validation (Jul–Dec 2025)

Test, Brier at 24 h: model **0.0177 [0.0146, 0.0213]** vs climatology baseline 0.0192 [0.0156, 0.0231] (paired difference −0.0014 [−0.0022, −0.0008]); monsoon months 0.0237 vs 0.0251. The model beats the declared baseline at 24, 48 and 72 h. Validation (used for model choice):

Brier [95% CI], lower is better	24 h	48 h	72 h
Climatology (declared baseline)	0.0249 [0.0188, 0.0320]	0.0249	0.0249
Raw forecast cut at threshold	0.0290 [0.0217, 0.0371]	0.0297	0.0294
Logistic regression	0.0238 [0.0182, 0.0305]	0.0244	0.0243
Gradient boosting (chosen)	**0.0233 [0.0177, 0.0297]**	**0.0236**	**0.0235**

Paired bootstrap, GBM − climatology at 24 h: −0.0016 [−0.0025, −0.0008]. The gain holds in monsoon months (skill +0.058) and in every province and terrain group; weakest: Azad Kashmir (+0.015). Buner, 15 Aug 2025: IMERG's wettest pixel (43 mm) stayed under Buner's threshold (57.5 mm), so the label is 0 despite the deadly flash flood; the model said 0.15, about 4× climatology. Model B did not beat "no change" (MAE 1.667 vs 1.665 pp).

Data

Source	Use
Open-Meteo Previous Runs (CC BY 4.0)	forecast features
NASA GPM IMERG Late V07	labels, thresholds, baseline
OCHA admin-2 boundaries (HDX)	160 districts
UNOSAT 2022 flood extent, Natural Earth, Copernicus DEM	district groups
PBS weekly SPI	Model B
NDMA sitreps and advisories (158 documents)	citations, local validation

Limits

- Predicts heavy rain, not floods or damage; not an official warning.
- Forecast inputs are taken at district centroids; labels use the wettest pixel.
- The run history misses part of the window (first run 01:28 PKT; laptop slept 04:33–06:10).
- Daily satellite rain smooths out local cloudbursts (the Buner case), so labels miss some deadly events.
- Urdu-only questions retrieve less well than English ones.

Impact

A free, district-level rain-risk signal that planners can check every hour, with the reasons (forecast rain against the district's own threshold) and official sources one click away.